

QUBO Formulation: Nurse Scheduling Assignment Problem

Nurse Scheduling Assignment Problem

Problem Statement. Given a set of N nurses indexed by $i \in \{0, \dots, N-1\}$ and a set of S shifts indexed by $j \in \{0, \dots, S-1\}$, let $c_{i,j} \in \mathbb{R}$ denote the cost of assigning nurse i to shift j . The goal is to assign exactly one nurse to each shift such that each nurse is assigned to at most one shift, while minimizing the total assignment cost.

Decision Variables. We introduce binary decision variables $x_{i,j} \in \{0, 1\}$ for all $i \in \{0, \dots, N-1\}$ and $j \in \{0, \dots, S-1\}$. The variable $x_{i,j}$ equals 1 if nurse i is assigned to shift j , and 0 otherwise. The total number of variables is $n = N \times S$. We map each pair (i, j) to a unique integer index $k = i \cdot S + j \in \{0, \dots, n-1\}$, identifying $x_{i,j}$ with the k -th variable in the QUBO vector.

QUBO Derivation. We construct the QUBO Hamiltonian $H(x)$ by combining the objective function with penalty terms that enforce the constraints.

Step 1 – Objective Function. The objective is to minimize the total cost:

$$\min_x f(x) = \sum_{i=0}^{N-1} \sum_{j=0}^{S-1} c_{i,j} x_{i,j}. \quad (1)$$

In the QUBO formulation, we minimize the energy directly, so the objective contribution is $H_{\text{obj}}(x) = f(x)$.

Step 2 – Constraints. The problem imposes two types of constraints:

1. *Shift coverage:* Each shift j must be covered by exactly one nurse:

$$\sum_{i=0}^{N-1} x_{i,j} = 1, \quad \forall j \in \{0, \dots, S-1\}. \quad (2)$$

2. *Nurse capacity:* Each nurse i can be assigned to at most one shift:

$$\sum_{j=0}^{S-1} x_{i,j} \leq 1, \quad \forall i \in \{0, \dots, N-1\}. \quad (3)$$

Since $x_{i,j}$ are binary, the inequality $\sum_j x_{i,j} \leq 1$ is equivalent to enforcing that no two distinct shifts are assigned to the same nurse. This can be expressed as the set of quadratic constraints:

$$x_{i,j}x_{i,k} = 0, \quad \forall i \in \{0, \dots, N-1\}, \forall 0 \leq j < k \leq S-1. \quad (4)$$

Step 3 – Penalty Terms. We introduce a penalty weight $\lambda > 0$ and convert constraints to penalty terms.

For the shift coverage constraints, we use the squared penalty:

$$P_j^{\text{shift}}(x) = \lambda \left(\sum_{i=0}^{N-1} x_{i,j} - 1 \right)^2. \quad (5)$$

Expanding using the idempotent property $x_{i,j}^2 = x_{i,j}$:

$$P_j^{\text{shift}}(x) = \lambda \left(\sum_{i=0}^{N-1} x_{i,j} + \sum_{i \neq i'} x_{i,j}x_{i',j} - 2 \sum_{i=0}^{N-1} x_{i,j} + 1 \right) \quad (6)$$

$$= \lambda \left(\sum_{i \neq i'} x_{i,j}x_{i',j} - \sum_{i=0}^{N-1} x_{i,j} + 1 \right), \quad (7)$$

where the sums over $i \neq i'$ run over distinct pairs of nurses for a fixed shift j .

For the nurse capacity constraints, we use the quadratic penalty derived from the pairwise products:

$$Q_i^{\text{nurse}}(x) = \lambda \sum_{0 \leq j < k \leq S-1} x_{i,j}x_{i,k}. \quad (8)$$

This term is zero if nurse i is assigned to at most one shift, and positive otherwise.

Step 4 – Combined Hamiltonian. The total QUBO Hamiltonian is the sum of the objective and all penalty terms:

$$H(x) = \sum_{i=0}^{N-1} \sum_{j=0}^{S-1} c_{i,j}x_{i,j} + \sum_{j=0}^{S-1} P_j^{\text{shift}}(x) + \sum_{i=0}^{N-1} Q_i^{\text{nurse}}(x). \quad (9)$$

Substituting the expanded forms:

$$H(x) = \sum_{i,j} c_{i,j}x_{i,j} + \lambda \sum_{j=0}^{S-1} \left(\sum_{i \neq i'} x_{i,j}x_{i',j} - \sum_{i=0}^{N-1} x_{i,j} + 1 \right) + \lambda \sum_{i=0}^{N-1} \sum_{0 \leq j < k \leq S-1} x_{i,j}x_{i,k}. \quad (10)$$

Step 5 – Q Matrix and Offset. We express $H(x)$ in the standard QUBO form $H(x) = x^\top Qx + c$, where x is the vector of variables indexed by $k = iS + j$. Collecting coefficients for linear and quadratic terms:

- *Diagonal entries:* The coefficient of $x_{i,j}$ comes from the objective and the shift penalty linear term:

$$Q_{k,k} = c_{i,j} - \lambda, \quad \text{where } k = iS + j. \quad (11)$$

- *Off-diagonal entries:* The coefficient of $x_{i,j}x_{i',j'}$ is non-zero only if the variables share a shift index or a nurse index.

- If $j = j'$ and $i \neq i'$ (same shift, different nurses), the term arises from the shift penalty quadratic part:

$$Q_{k,k'} = \lambda, \quad \text{where } k = iS + j, \quad k' = i'S + j. \quad (12)$$

- If $i = i'$ and $j < j'$ (same nurse, different shifts), the term arises from the nurse penalty:

$$Q_{k,k'} = \lambda, \quad \text{where } k = iS + j, \quad k' = iS + j'. \quad (13)$$

- Otherwise, $Q_{k,k'} = 0$.

- *Offset:* The constant term arises solely from the shift penalties:

$$c = \sum_{j=0}^{S-1} \lambda \cdot 1 = S\lambda. \quad (14)$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any infeasible solution has a strictly higher energy than any feasible solution. The minimum penalty incurred by violating a shift coverage constraint is λ (e.g., if a shift is assigned to zero or two or more nurses). The maximum possible reduction in the objective function value relative to a feasible assignment is bounded by the sum of the absolute costs of all possible assignments. Therefore, choosing

$$\lambda > \sum_{i=0}^{N-1} \sum_{j=0}^{S-1} |c_{i,j}| \quad (15)$$

guarantees that the penalty dominates any potential objective gain from violating constraints, ensuring that the global minimum of $H(x)$ corresponds to a feasible solution.

Correctness Argument. Minimizing $H(x)$ over $x \in \{0, 1\}^n$ recovers the optimal schedule as follows. For any feasible solution x^{feas} , all constraints are satisfied, so all penalty terms vanish, and $H(x^{\text{feas}}) = f(x^{\text{feas}})$. For any infeasible solution x^{infeas} , at least one constraint is violated, incurring a penalty of at least λ . By the choice of λ , we have $H(x^{\text{infeas}}) \geq \lambda > \sum |c_{i,j}| \geq f(x^{\text{feas}})$ for any feasible x^{feas} . Thus, $H(x^{\text{infeas}}) > H(x^{\text{feas}})$ for all feasible x^{feas} . Consequently, the global minimum of $H(x)$ must be attained at a feasible solution, and among feasible solutions, H equals the objective function, so the minimum corresponds to the optimal assignment.

Illustrative Example. Consider a small instance with $N = 3$ nurses and $S = 3$ shifts, yielding $n = 9$ variables. The variables are indexed $k = 0, \dots, 8$ with mapping $x_{i,j} \leftrightarrow x_k$ where $k = 3i + j$. The cost matrix entries are denoted c_{ij} .

The QUBO Hamiltonian is:

$$\begin{aligned} H(x) = & \sum_{i=0}^2 \sum_{j=0}^2 c_{ij} x_{3i+j} \\ & + \lambda \sum_{j=0}^2 \left(\sum_{i \neq i'} x_{3i+j} x_{3i'+j} - \sum_{i=0}^2 x_{3i+j} + 1 \right) \\ & + \lambda \sum_{i=0}^2 (x_{3i} x_{3i+1} + x_{3i} x_{3i+2} + x_{3i+1} x_{3i+2}). \end{aligned} \quad (16)$$

Expanding the shift penalties for each j :

$$\sum_{j=0}^2 (x_{3i+j} x_{3i'+j}) = (x_0 x_3 + x_0 x_6 + x_3 x_6) + (x_1 x_4 + x_1 x_7 + x_4 x_7) + (x_2 x_5 + x_2 x_8 + x_5 x_8), \quad (17)$$

and the linear shift terms:

$$- \sum_{j=0}^2 \sum_{i=0}^2 x_{3i+j} = -(x_0 + x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8). \quad (18)$$

The nurse capacity terms are:

$$\sum_{i=0}^2 \sum_{0 \leq j < k \leq 2} x_{3i+j} x_{3i+k} = (x_0 x_1 + x_0 x_2 + x_1 x_2) + (x_3 x_4 + x_3 x_5 + x_4 x_5) + (x_6 x_7 + x_6 x_8 + x_7 x_8). \quad (19)$$

Collecting terms, the QUBO takes the form $H(x) = x^\top Q x + c$ with:

$$Q_{k,k} = c_{ij} - \lambda, \quad \text{for } k = 3i + j, \ i, j \in \{0, 1, 2\}, \quad (20)$$

$$Q_{k,k'} = \lambda \quad \text{if } (x_k, x_{k'}) \text{ share a shift index or a nurse index and } k \neq k', \quad (21)$$

and all other off-diagonal entries are zero. The offset is $c = 3\lambda$.

For instance, the linear coefficient for x_0 (nurse 0, shift 0) is $c_{00} - \lambda$. The quadratic coefficient for $x_0 x_3$ (nurses 0 and 1 on shift 0) is λ . The quadratic coefficient for $x_0 x_1$ (nurse 0 on shifts 0 and 1) is λ . All other pairs have coefficient zero. The penalty weight must satisfy $\lambda > \sum_{i=0}^2 \sum_{j=0}^2 |c_{ij}|$.